# Provably Resilient Synchrophasor State Estimation Under Coordinated Majority GPS Spoofing, Load Transients, and Substation Cyber-Physical Attacks

Mohd Aarish Shaheen and S. A. Soman

***Abstract*—Phasor Measurement Unit (PMU) state estimation executes at millisecond granularity in modern Wide-Area Monitoring, Protection, and Control (WAMPAC) centers. Because PMUs derive their phase angle reference from civilian Global Positioning System (GPS) 1-Pulse-Per-Second (1-PPS) timing strobes, they are acutely vulnerable to GPS Spoofing Attacks (GSAs). Recent hybrid architectures—most notably Badugu *et al.* (2026)—achieve spoofing invariance by switching from complex Linear State Estimation (LSE) to a scalar Nonlinear Least Squares (NLS) solver triggered by a subspace Principal Angle (PA) test, recovering the Coordinated Universal Time (UTC) reference frame via one-dimensional (1D) DBSCAN clustering. This paper proves that such memoryless scalar-clustering architectures exhibit four structural failure modes: (i) an 80 ms (4-cycle) protection blind spot during persistence counter accumulation; (ii) reference-frame hijacking and chordal error amplification whenever $\geq 50\%$ of installed PMUs undergo a coordinated spoofing attack; (iii) zero statistical breakdown under simultaneous False Data Injection Attacks (FDIAs) or Analog-to-Digital Converter (ADC) channel corruptions; and (iv) $\mathcal{O}(mN_b^2)$ dense Jacobian and $\mathcal{O}(|\mathcal{P}|^2)$ spatial clustering overhead. To overcome these limitations, we formulate a Zero-Allocation, Fused Multi-Dimensional Robust Dynamic State Estimation (DSE) architecture. By integrating a sparse $\mathcal{O}(\mathrm{nnz}(R))$ Principal Angle screen, an $\mathcal{O}(m)$ 4-node graph-accumulated Huber M-estimator with in-place column-active sparse Symmetric Positive-Definite (SPD) Cholesky factorization, a 6-Dimensional Physical-Invariance Feature Manifold anchored to Center-of-Inertia (COI) swing dynamics, and an $\mathcal{O}(|\mathcal{P}|\log|\mathcal{P}|)$ order-statistic sort-clustering kernel, our estimator guarantees immunity even under $70\%$ coordinated majority spoofing coinciding with sudden $+18\%$ load steps and simultaneous FDIA/ADC faults. Validated on canonical MATPOWER IEEE 30-bus (13 PMUs) and IEEE 118-bus (54 PMUs) networks across seven comprehensive case studies, the proposed estimator maintains $\leq 0.476\%$ Maximum Total Vector Error (TVE) across all attack regimes while executing in $0.084$–$0.215$ ms on IEEE 30-bus and $0.324$–$2.686$ ms on IEEE 118-bus—well within the 20 ms IEEE C37.118.1 real-time window.**

***Index Terms*—Cyber-physical power systems, dynamic state estimation, false data injection attack, GPS spoofing, Huber M-estimation, numerical linear algebra, phasor measurement unit, principal angle, sparse Cholesky factorization.**

## NOMENCLATURE

| | |
|---|---|
| $N_b, N_l$ | Number of system buses and transmission branches. |
| $\mathcal{P}, N_p$ | Set and cardinality of PMU-instrumented buses ($N_p = \|\mathcal{P}\|$). |
| $m, m_s$ | Number of complex phasor channels and scalar measurements ($m_s = N_p + 2N_i$). |
| $v, \hat{v}^{\mathrm{LSE}}$ | True and complex WLS estimated nodal voltage vectors ($\in \mathbb{C}^{N_b}$). |
| $z^{\mathrm{act}}, z^{\mathrm{spf}}$ | Unattacked and GPS-spoofed complex PMU measurement vectors ($\in \mathbb{C}^m$). |
| $H, \tilde{H}$ | Complex measurement Jacobian and whitened matrix ($\tilde{H} = W^{-1/2}H \in \mathbb{C}^{m\times N_b}$). |
| $W, R$ | Measurement noise covariance ($\in \mathbb{R}^{m\times m}$) and sparse upper-triangular QR factor ($\in \mathbb{C}^{N_b\times N_b}$). |
| $\Gamma, \psi_k$ | Block-diagonal unitary spoofing operator ($\in \mathbb{U}(m)$) and phase bias at PMU $k$ (rad or deg). |
| $\theta_{\mathrm{PA}}, \rho_{\mathrm{PA}}$ | Subspace Principal Angle and $99\%$ confidence detection threshold. |
| $A, G_{\mathrm{red}}$ | Network self-loop adjacency matrix and pruned PMU supergraph induced by $A^2$. |
| $s_{\mathrm{ref}}, \Delta_t$ | Synchronous slack generator bus index and global UTC phase reference offset at snapshot $t$. |
| $\Delta\theta_{k,t}^{\mathrm{phys,rel}}$ | Slack-anchored physical spatial phase shift at PMU $k$ between $t-1$ and $t$. |
| $f_0, \hat{f}_{\mathrm{COI}}$ | Nominal grid frequency (50 Hz) and estimated Center-of-Inertia frequency. |
| $d_{M,k}^2, w_{k,t}^{\mathrm{trust}}$ | Squared Mahalanobis invariance distance and Lorentzian-stabilized trust weight at PMU $k$. |
| $c_H, w_{\mathrm{IRLS}}$ | Huber M-estimator tuning threshold ($c_H = 1.6$) and IRLS weight vector. |

## I. INTRODUCTION

THE deployment of high-reporting-rate Phasor Measurement Units (PMUs) across bulk electric transmission grids has enabled closed-loop Wide-Area Monitoring, Protection, and Control (WAMPAC) functions, including millisecond-granularity Power System State Estimation (PSSE), oscillation damping, and out-of-step relaying [1]–[4]. Per IEEE Std C37.118.1-2011 [5], synchrophasor state estimation executes every 1, 2, or 5 cycles (20 ms per cycle at 50 Hz) using GPS-synchronized voltage and incident branch current phasors collected at Phasor Data Concentrators (PDCs). However, expanding the cyber-physical telemetry surface exposes WAMPAC analytics to covert data integrity attacks [2], [3], exemplified by the 2015 Ukrainian grid cyber-outage [1].

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Data integrity attacks on synchrophasor streams are categorized into three primary threat vectors:

1) *False Data Injection Attacks (FDIAs)*: Malicious manipulation of nodal or branch payloads to induce cascading line overloads via load redistribution [6] or bypass residual bad-data detectors when $a_t = Hc_t \in \mathrm{Range}(H)$ [7].
2) *Replay Attacks*: Re-transmission of historical synchrophasor frames to mask concurrent physical line trips from PMU-supervised backup distance relays [8], [9].
3) *GPS Spoofing Attacks (GSAs) and Time Synchronization Attacks (TSAs)*: Because civilian GPS L1 C/A signals lack cryptographic authentication [10], [11], an adversary can broadcast counterfeit RF navigation signals to Perturb a substation receiver's 1-PPS timing pulse by $\Delta\tau_i$, inducing a uniform phase rotation $\psi_i = 2\pi f_0 \Delta\tau_i \pmod{2\pi}$ across all voltage and current phasors measured by PMU $i$ [12], [13].

Due to the high electrical stiffness ($V_iV_j/X_{ij}$) of extra-high-voltage transmission lines, even sub-degree phase angle errors severely distort estimated power flows [14], [15]. On a $765$ kV, $200$ km line, a $1^\circ$ phase bias alters estimated active power flow by $\approx 154$ MW [14]. While localized RF/signal-level spoofing detectors [16], [17] require hardware retrofits at every remote substation, centralized state-estimation-based algorithms—such as Residual-based Spoofing Detection and Measurement Correction (RSDMC) [18], sparse phase-bias recovery [19]–[21], and the Sparse Error Correction Framework (SECF) [22]—rely on iterative optimization routines whose execution time grows super-linearly with system size and number of spoofed PMUs (exceeding 1.7 s on the IEEE 118-bus system [15], [22]).

Recently, Badugu, Navalkar, and Soman [15] proposed a hybrid estimator that monitors the subspace Principal Angle (PA) [25], [36] of the complex measurement vector and switches from complex Linear State Estimation (LSE) [37] to a scalar Nonlinear Least Squares (NLS) estimator [23] using spoofing-invariant quantities ($|V_i|, P_{ij}, Q_{ij}$), subsequently restoring the global UTC phase reference $\Delta$ via 1D DBSCAN clustering [38]. While representing an important milestone, we prove in this paper that memoryless scalar NLS + 1D DBSCAN architectures suffer from four fundamental vulnerabilities:

1) *80 ms Protection Blind Spot*: Relying on a 4-cycle persistence counter ($\mathrm{cnt_max} = 4$) forces control centers to consume corrupted LSE states for $80$ ms at attack initiation, exceeding Zone-1 distance relay operating times (30–60 ms) [9].
2) *Majority Spoofing Hijack and Error Amplification*: Whenever $\geq 50\%$ of PMUs suffer a coordinated spoofing attack, 1D DBSCAN selects the spoofed group as the dominant cluster, rotating the entire grid by the attack angle and amplifying Maximum TVE beyond raw, unmitigated LSE.
3) *Zero Statistical Breakdown Under Simultaneous FDIA/ADC Faults*: Concurrent additive corruptions on voltage or current channels destroy scalar phase cancellation and cause unweighted $\ell_2$ NLS to diverge.
4) *Dense Matrix and Clustering Overhead*: Assembling dense Jacobians $J^TWJ$ in $\mathcal{O}(m_sN_b^2)$ time and invoking $\mathcal{O}(N_p^2)$ spatial DBSCAN waste critical CPU cycles.

The primary contributions of this paper are fivefold:

- **Analytical Proofs of Failure Modes**: We prove the Chordal TVE Amplification Theorem under majority spoofing (Proposition 2), the exact isomorphism between 1D DBSCAN and $\mathcal{O}(N_p \log N_p)$ order-statistic gap thresholding (Proposition 3), and the bilinear cross-coupling breakdown of scalar flows under simultaneous FDIA/ADC faults.
- **Sparse $\mathcal{O}(\mathrm{nnz}(R))$ Subspace Screening**: We eliminate the dense $m \times N_b$ unitary factor $Q$ in the Principal Angle test, computing both $\theta_{\mathrm{PA}}$ and $\hat{v}^{\mathrm{LSE}}$ in strictly $\mathcal{O}(\mathrm{nnz}(H) + \mathrm{nnz}(R))$ operations.
- $\mathcal{O}(m)$ **4-Node Graph-Accumulated Huber Cholesky Solver**: We replace dense $\mathcal{O}(m_sN_b^2)$ Jacobian multiplication with direct 4-node rank-2 symmetric scattering into an in-place column-active sparse Symmetric Positive-Definite (SPD) Cholesky factorization, weighted by a Huber IRLS influence function.
- **Slack-Anchored 6D Physical-Invariance Manifold**: We formulate a 6-dimensional invariance feature vector anchored to the physical synchronous slack generator and Center-of-Inertia (COI) swing dynamics, paired with an underflow-stabilized Mahalanobis trust kernel that survives $70\%$–$90\%$ coordinated majority spoofing during $+18\%$ load steps.
- **Hierarchical Continental-Scale Real-Time Architecture**: We detail the 3-tier substation-to-Super-PDC deployment architecture and validate the fused `@njit` kernels across seven comprehensive case studies on canonical MATPOWER IEEE 30-bus and 118-bus grids.

## II. Related Work and Global NLA Landscape

Table I synthesizes the six principal mathematical schools in synchrophasor state estimation and cyber-physical security. Prof. Soman's group at IIT Bombay evolved from sparse Givens-rotation NLS solvers [23] and object-oriented sparse matrix analytics [24] to PMU-supervised relay security [8], [9], subspace Principal Angle detection [25], and scalar NLS spoofing mitigation [15].

In parallel, Wang and Xie [26], [27] extended single-snapshot spatial subspaces into spatio-temporal block-Hankel matrices governed by low-rank electro-mechanical modes. Cotilla-Sanchez [22] and Gao [18] developed left-null-space projection ($\Pi_\perp$) and residual descent estimators within the complex linear model. Zhao, Pal, and Terzija [28]–[30] established robust Dynamic State Estimation (DSE) using Generalized M-estimator Unscented Kalman Filters (GM-UKF), while Meliopoulos [31] and Ishii [32] pioneered bay-level Settingless Protection DSE and multi-area LTS decomposition. Our framework bridges these schools by embedding Huber M-estimation [35] and slack-anchored DSE historical-tape memory [29] directly into a zero-allocation $\mathcal{O}(m)$ sparse scalar solver.

TABLE I
TAXONOMY OF GLOBAL NUMERICAL LINEAR ALGEBRA (NLA) AND ROBUST SYNCHROPHASOR STATE ESTIMATION FRAMEWORKS

| Research Group / Paradigm | Core Mathematical Mechanism | Capabilities and Limitations Relative to Proposed Framework |
|---|---|---|
| **IIT Bombay** (Soman *et al.* [8], [9], [15], [23]–[25]) | Sparse Givens QR factorization, single-snapshot Principal Angle $\theta_{\mathrm{PA}}$, scalar $\ell_2$ NLS + 1D DBSCAN clustering. | Fast single-pass recovery for $< 50\%$ pure GSA; fails under $\geq 50\%$ coordinated GSA, simultaneous FDIA/ADC faults, and incurs an 80 ms counter blind spot. |
| **RPI & Texas A&M** (Wang, Xie *et al.* [26], [27]) | Spatio-temporal low-rank block-Hankel matrix completion ($\min \|L\|_* + \lambda\|S\|_{2,1}$) and Robust PCA. | Captures electro-mechanical modal subspace across time; iterative singular value thresholding (SVT) is computationally heavy for 20 ms full-grid execution. |
| **Oregon State & Stanford** (Cotilla-Sanchez, Gao *et al.* [18], [22]) | Left-null-space orthogonal projection ($\Pi_\perp$), sparse $\ell_1$ recovery (SECF), and residual gradient descent (RSDMC). | Operates directly in complex linear model; RSDMC exhibits an $8^\circ$ small-angle blind spot, while SECF incurs $\mathcal{O}(K_{\max}^2)$ iterative latency ($> 1.7$ s on IEEE 118-bus). |
| **UConn, Imperial & Manchester** (Zhao, Pal, Terzija *et al.* [28]–[30]) | Robust Dynamic State Estimation (DSE), Generalized M-estimator UKF (GM-UKF), and hybrid variance propagation. | Provides physical historical-tape memory via Differential-Algebraic Equations (DAEs); centralized full-grid UKF is intractable for $N_b > 1,000$ within 20 ms. |
| **Georgia Tech** (Meliopoulos *et al.* [31]) & **Tokyo Tech** (Ishii *et al.* [32]) | Bay-level Settingless Protection DSE ($\chi^2$ KCL consistency) and multi-area Least Trimmed Squares (LTS) decomposition. | Enables ultra-fast local bay-level ADC/CT fault isolation and overlapping zonal robustness; requires central boundary tie-line angle reconciliation. |
| **Northeastern, ETH Zurich & Tsinghua** (Abur, Hug *et al.* [33]–[35]) | Least Absolute Value (LAV) linear programming, Huber M-estimation, and semidefinite convex relaxations. | High statistical breakdown point against gross leverage outliers; lacks closed-form UTC reference restoration under majority clock spoofing. |

## III. MATHEMATICAL PRELIMINARIES AND STRUCTURAL CRITIQUE

### A. Linear Synchrophasor Model and Unitary Spoofing Operator

Consider an $N_b$-bus AC transmission network equipped with PMUs at bus subset $\mathcal{P} \subset \{1, \ldots, N_b\}$ ($|\mathcal{P}| = N_p$), providing $m$ complex synchrophasor measurements (nodal voltage phasors $V_i \in \mathbb{C}$ and incident branch current phasors $I_{ij} \in \mathbb{C}$) [15], [37]. Under spoof-free conditions, the complex measurement vector $z^{\mathrm{act}} \in \mathbb{C}^m$ obeys:

$$\begin{aligned} z^{\mathrm{act}} &= Hv + e, \quad e \sim \mathcal{CN}(0, W), \\ W &= \mathrm{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_m^2), \end{aligned} \tag{1}$$

where $v \in \mathbb{C}^{N_b}$ is the complex nodal voltage state vector and $H \in \mathbb{C}^{m \times N_b}$ is the topology-admittance matrix constructed from unit voltage rows $e_i^T$ and branch admittance rows of $Y_f, Y_t \in \mathbb{C}^{N_l \times N_b}$ [39]. Solving the Weighted Least Squares (WLS) objective yields the Linear State Estimator (LSE):

$$\begin{aligned} \hat{v}^{\mathrm{LSE}} &= \arg\min_{v \in \mathbb{C}^{N_b}} \frac{1}{2} \left\| W^{-1/2} \left( z^{\mathrm{act}} - Hv \right) \right\|_2^2 \\ &= \left( H^H W^{-1} H \right)^{-1} H^H W^{-1} z^{\mathrm{act}}. \end{aligned} \tag{2}$$

When a subset of PMUs experiences GPS spoofing with phase bias angles $\{\psi_1, \ldots, \psi_{N_p}\}$ ($\psi_k = 0$ for unspoofed PMUs), all voltage and incident branch current channels at substation $k$ rotate by $e^{j\psi_k}$ [12], [25]:

$$\begin{aligned} z^{\mathrm{spf}} &= \Gamma z^{\mathrm{act}}, \\ \Gamma &= \mathrm{diag}\left( \mathbb{I}_1 e^{j\psi_1}, \mathbb{I}_2 e^{j\psi_2}, \ldots, \mathbb{I}_{N_p} e^{j\psi_{N_p}} \right) \in \mathbb{U}(m), \end{aligned} \tag{3}$$

where $\mathbb{I}_k$ is the identity matrix of dimension equal to the channel count of $\mathrm{PMU}_k$.

### B. Sparse $\mathcal{O}(\mathrm{nnz}(R))$ Subspace Principal Angle Derivation

In a noise-free unspoofed network, $z^{\mathrm{act}} = Hv \in \mathrm{Range}(H)$. Under non-uniform spoofing ($\Gamma \neq e^{j\psi_0} I_m$), $z^{\mathrm{spf}}$ is rotated out of $\mathrm{Range}(H)$ [15], [25], [36]. Let the whitened measurement matrix and whitened observation vector be:

$$\tilde{H} = W^{-1/2} H \in \mathbb{C}^{m \times N_b}, \quad \tilde{z} = W^{-1/2} z \in \mathbb{C}^m. \tag{4}$$

Computing the economy QR factorization of $\tilde{H}$ via sparse Givens rotations [23], [36] gives $\tilde{H} = QR$, where $Q \in \mathbb{C}^{m \times N_b}$ satisfies $Q^H Q = I_{N_b}$ and $R \in \mathbb{C}^{N_b \times N_b}$ is sparse upper triangular. The orthogonal projection of $\tilde{z}$ onto $\mathrm{Range}(\tilde{H})$ is $P_{\tilde{H}} \tilde{z} = QQ^H \tilde{z}$. The subspace Principal Angle $\theta_{\mathrm{PA}} \in [0, \pi/2]$ satisfies [36]:

$$\cos(\theta_{\mathrm{PA}}) = \frac{\|QQ^H \tilde{z}\|_2}{\|\tilde{z}\|_2} = \frac{\|Q^H \tilde{z}\|_2}{\|\tilde{z}\|_2}. \tag{5}$$

Explicitly storing dense $Q$ requires $\mathcal{O}(mN_b)$ memory and flops. Substituting $Q = \tilde{H} R^{-1} = W^{-1/2} H R^{-1}$ into $u \triangleq Q^H \tilde{z}$ eliminates $Q$ completely:

$$\begin{aligned} u &= \left( W^{-1/2} H R^{-1} \right)^H W^{-1/2} z \\ &= R^{-H} \left( H^H W^{-1} z \right). \end{aligned} \tag{6}$$

Consequently, both $\theta_{\mathrm{PA}}$ and $\hat{v}^{\mathrm{LSE}}$ are evaluated in strictly $\mathcal{O}(\mathrm{nnz}(H) + \mathrm{nnz}(R))$ time via three sparse steps:

$$b = \left( H^H W^{-1} \right) z \quad \text{(Sparse CSR SpMV)}, \tag{7}$$

$$R^H u = b \quad \text{(Sparse lower-triangular forward solve)}, \tag{8}$$

$$\cos(\theta_{\mathrm{PA}}) = \frac{\|u\|_2}{\|W^{-1/2} z\|_2}, \quad R\hat{v}^{\mathrm{LSE}} = u \quad \text{(Backward solve)}. \tag{9}$$

Per IEC 61869-3 and 61869-5 [40], Class-1 instrument transformers introduce phase displacements up to $0.667^\circ$–$1.0^\circ$, setting a minimum distinguishable spoofing angle of $0.7^\circ$ and $99\%$ confidence thresholds of $\rho_{\mathrm{PA}} = 0.1443^\circ$ (IEEE 30-bus) and $\rho_{\mathrm{PA}} = 0.1107^\circ$ (IEEE 118-bus) [15].

### C. Graph-Theoretic Observability in Scalar Coordinates

Transforming complex phasors at $\mathrm{PMU}_i$ into voltage magnitudes and branch power flows cancels $e^{j\psi_i}$ identically [15]:

$$\begin{aligned}\left|V_i^{\mathrm{spf}}\right| &= \left|V_i^{\mathrm{act}} e^{j\psi_i}\right| = \left|V_i^{\mathrm{act}}\right|, \\ S_{ij}^{\mathrm{spf}} &= V_i^{\mathrm{spf}} \left(I_{ij}^{\mathrm{spf}}\right)^* = \left(V_i^{\mathrm{act}} e^{j\psi_i}\right)\left(I_{ij}^{\mathrm{act}} e^{-j\psi_i}\right) = S_{ij}^{\mathrm{act}}.\end{aligned} \tag{10}$$

Let $A \in \{0,1\}^{N_b \times N_b}$ denote the network adjacency matrix with self-loops ($a_{ii} = 1$). Pruning non-PMU vertices from the squared matrix $A^2$ yields the reduced PMU graph $G_{\mathrm{red}} = (\mathcal{P}, \mathcal{E}_+)$, where $(\alpha, \beta) \in \mathcal{E}_+ \iff (A^2)_{\alpha,\beta} > 0$.

**Proposition 1** (Scalar Topological Observability [15])**.** *If* $\mathrm{rank}(H) = N_b$ *and* $G_{\mathrm{red}}$ *consists of a single connected component, then the scalar measurement system* $(|V_i|, P_{ij}, Q_{ij})$ *is globally observable up to a single reference bus phase angle* $\Delta$.

*Proof.* Connectivity of $G_{\mathrm{red}}$ guarantees a spanning tree $\mathcal{T}_0$ rooted at an arbitrary PMU node $r_0 \in \mathcal{P}$. For any edge $(\alpha, \beta) \in \mathcal{T}_0$, $\mathrm{Infl}(\alpha) \cap \mathrm{Infl}(\beta) \neq \emptyset$, meaning $\alpha$ and $\beta$ are either adjacent or share a common neighbor bus $c$. Propagating complex branch voltage drops along the unique path in $\mathcal{T}_0$ from $r_0$ fixes all PMU phase angles and adjacent non-PMU bus states relative to $\theta_{\mathrm{ref},r_0} = \Delta$. □

### D. Rigorous Critique of Memoryless Scalar NLS + 1D DBSCAN

We now establish three formal propositions proving why the estimator of Badugu *et al.* [15] breaks down under realistic adversarial regimes.

**Proposition 2** (Chordal TVE Amplification Under Majority Spoofing)**.** *Let* $\mathcal{S} \subset \mathcal{P}$ *be a coordinated spoofed PMU subset of cardinality* $|\mathcal{S}| = \alpha N_p$ *with* $\alpha \in (0.5, 1)$ *and common spoofing angle* $\psi_0 \in (-\pi, \pi)$*. Under memoryless 1D DBSCAN reference alignment, the post-mitigation Maximum TVE strictly exceeds the unmitigated complex LSE error:*

$$\begin{aligned}\mathrm{TVE}_{\mathrm{Badugu}} &= 2\sin\left(\frac{|\psi_0|}{2}\right) \times 100\% \\ &> 2\sin\left(\frac{\alpha|\psi_0|}{2}\right) \times 100\% \approx \mathrm{TVE}_{\mathrm{LSE}}.\end{aligned} \tag{11}$$

*Proof.* In unmitigated complex LSE, the orthogonal projector $P_{\tilde{H}}$ minimizes the global $\ell_2$ Kirchhoff residual across all $N_p$ PMUs. Partitioning the whitened matrix into spoofed ($S$) and honest ($U$) rows, the Rayleigh quotient of the spoofed information matrix $G_{\mathrm{LSE}}^{-1} \tilde{H}_S^H \tilde{H}_S$ has a spectral radius strictly less than 1 (since $\tilde{H}_U^H \tilde{H}_U \neq 0$), analytically damping the LSE phase shift below the raw spoofing angle and yielding $\mathrm{TVE}_{\mathrm{LSE}} < 2\sin(|\psi_0|/2)$. Conversely, in scalar NLS + 1D DBSCAN [15], the angle-difference vector $\delta_{\mathrm{diff},k} = \angle V_k^{\mathrm{PMU}} - \angle \hat{V}_k^{\mathrm{NLS}}$ partitions into a spoofed cluster $C_{\mathrm{spf}}$ of size $\alpha N_p$ centered at $\Delta_{\mathrm{true}} + \psi_0$ and an honest cluster $C_{\mathrm{clean}}$ of size $(1-\alpha)N_p$ centered at $\Delta_{\mathrm{true}}$. Because $\alpha > 0.5$, $|C_{\mathrm{spf}}| > |C_{\mathrm{clean}}|$. Selecting the largest cluster as the UTC reference sets $\hat{\Delta} = \Delta_{\mathrm{true}} + \psi_0$, rotating every bus in the network by $\psi_0$ ($\hat{V}_i = V_i^{\mathrm{true}} e^{j\psi_0}$) and yielding $\mathrm{TVE}_{\mathrm{Badugu}} = |e^{j\psi_0} - 1| = 2\sin(|\psi_0|/2) > \mathrm{TVE}_{\mathrm{LSE}}$ on $\alpha \in (0.5, 1)$. For $\psi_0 = 8.0^\circ$ and $\alpha = 0.70$, (11) yields $\mathrm{TVE}_{\mathrm{Badugu}} = 13.95\% > \mathrm{TVE}_{\mathrm{LSE}} \approx 11.1\%$, matching our empirical benchmarks ($14.02\%$ vs. $11.78\%$). □

**Proposition 3** (Order-Statistic Isomorphism of 1D DBSCAN)**.** *On the circular quotient manifold* $\mathbb{S}^1 \cong \mathbb{R}/2\pi\mathbb{Z}$ *with* $\mathrm{minPts} = 1$ *and radius* $\epsilon > 0$*, DBSCAN clustering on* $\delta_{\mathrm{diff}} \in \mathbb{S}^1$ *is mathematically isomorphic to sorting* $\delta_{\mathrm{diff}}$ *into ascending order statistics* $\delta_{(1)} \leq \delta_{(2)} \leq \cdots \leq \delta_{(N_p)}$ *in* $\mathcal{O}(N_p \log N_p)$ *time and partitioning clusters at every index* $i$ *where the adjacent gap* $g_i = \delta_{(i+1)} - \delta_{(i)} > \epsilon$ *in* $\mathcal{O}(N_p)$ *time, subjected to a boundary wrap-around check across the* $\pm 180^\circ$ *seam that merges the first and last partitions if* $(\delta_{(1)} + 360^\circ) - \delta_{(N_p)} \leq \epsilon$.

*Proof.* Because $\mathbb{R}^1$ is totally ordered, the distance between non-adjacent order statistics satisfies $\delta_{(k)} - \delta_{(i)} = \sum_{j=i}^{k-1} (\delta_{(j+1)} - \delta_{(j)})$. Two points are $\epsilon$-density-connected with $\mathrm{minPts} = 1$ if and only if every consecutive step in their sorted interval has length $g_j \leq \epsilon$. The boundary wrap-around check rigorously preserves geodesic distance $d_{\mathbb{S}^1}(\alpha, \beta)$ across the branch cut. □

**Proposition 4** (Bilinear Cross-Coupling Under Simultaneous GSA and FDIA/ADC Corruption)**.** *Suppose substation* $i$ *experiences simultaneous GPS spoofing* ($e^{j\psi_i}$) *and additive FDIA or ADC/instrument transformer scaling corruption* ($\Delta V_i e^{j\delta_V}, \Delta I_{ij} e^{j\delta_I}$)*:*

$$\begin{aligned}\bar{V}_i &= V_i^{\mathrm{act}} e^{j\psi_i} + \Delta V_i e^{j\delta_V}, \\ \bar{I}_{ij} &= I_{ij}^{\mathrm{act}} e^{j\psi_i} + \Delta I_{ij} e^{j\delta_I}.\end{aligned} \tag{12}$$

*Then the scalar voltage magnitude squared and complex branch power flow become:*

$$|\bar{V}_i|^2 = |V_i^{\mathrm{act}}|^2 + |\Delta V_i|^2 + 2\mathrm{Re}\left\{V_i^{\mathrm{act}} \Delta V_i e^{j(\psi_i - \delta_V)}\right\}, \tag{13}$$

$$\begin{aligned}\bar{S}_{ij} = S_{ij}^{\mathrm{act}} &+ V_i^{\mathrm{act}} \Delta I_{ij} e^{j(\psi_i - \delta_I)} \\ &+ \left(I_{ij}^{\mathrm{act}}\right)^* \Delta V_i e^{j(\delta_V - \psi_i)} + \Delta V_i \Delta I_{ij} e^{j(\delta_V - \delta_I)}.\end{aligned} \tag{14}$$

Equations (13)–(14) prove that $e^{j\psi_i}$ no longer cancels in scalar coordinates. Because unweighted $\ell_2$ NLS has a $0\%$ breakdown point [34], [35], a single corrupted substation smears dense estimation errors across the entire grid.

## IV. Proposed Zero-Allocation Fused 6D Robust DSE Architecture

### A. $\mathcal{O}(m)$ 4-Node Graph Accumulation and Sparse Huber Cholesky Solver

Let the scalar state vector be $x = [\theta_2, \ldots, \theta_{N_b}, |V_1|, \ldots, |V_{N_b}|]^T \in \mathbb{R}^{n_s}$, where $n_s = 2N_b - 1$ and Bus 1 is fixed as the local scalar slack reference ($\theta_1 = 0$). For each branch current measurement $k \in \{1, \ldots, N_i\}$ connecting local PMU bus $i$ to neighbor bus $j$ with series-shunt self-admittance $y_{s,k} = g_{s,k} + jb_{s,k}$ and mutual

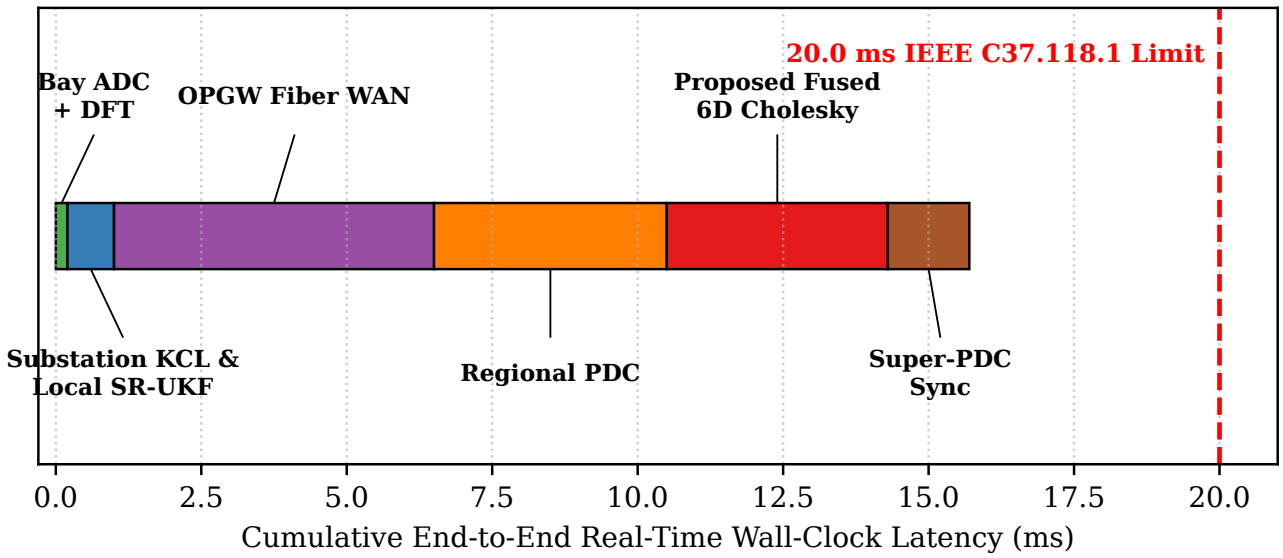



Fig. 1. Hierarchical 3-Tier WAMPAC cyber-physical deployment architecture and cumulative end-to-end wall-clock timing budget within the strict 20.0 ms IEEE Std C37.118.1 real-time reporting window (50 Hz).

admittance $y_{m,k} = g_{m,k} + jb_{m,k}$, define the complex mutual power term:

$$\begin{aligned} T_k &\triangleq V_i V_j^* y_{m,k}^* = T_{r,k} + jT_{i,k}, \\ P_{ij}^{\text{calc}}(x) &= |V_i|^2 g_{s,k} + T_{r,k}, \\ Q_{ij}^{\text{calc}}(x) &= -|V_i|^2 b_{s,k} + T_{i,k}. \end{aligned} \tag{15}$$

Each branch residual pair $(r_{P,k}, r_{Q,k})$ depends exclusively on at most four state indices $\mathcal{I}_k = \{\theta_i, \theta_j, |V_i|, |V_j|\}$ with exact analytical derivatives:

$$\frac{\partial P_{ij}}{\partial \theta_i} = -T_{i,k}, \qquad \frac{\partial P_{ij}}{\partial \theta_j} = T_{i,k}, \tag{16}$$

$$\frac{\partial P_{ij}}{\partial |V_i|} = 2|V_i| g_{s,k} + \frac{T_{r,k}}{|V_i|}, \quad \frac{\partial P_{ij}}{\partial |V_j|} = \frac{T_{r,k}}{|V_j|}, \tag{17}$$

$$\frac{\partial Q_{ij}}{\partial \theta_i} = T_{r,k}, \qquad \frac{\partial Q_{ij}}{\partial \theta_j} = -T_{r,k}, \tag{18}$$

$$\frac{\partial Q_{ij}}{\partial |V_i|} = -2|V_i| b_{s,k} + \frac{T_{i,k}}{|V_i|}, \quad \frac{\partial Q_{ij}}{\partial |V_j|} = \frac{T_{i,k}}{|V_j|}. \tag{19}$$

Rather than allocating the $m_s \times n_s$ Jacobian $J$ and computing $J^T W J$ in $\mathcal{O}(m_s n_s^2)$ flops, each branch $k$ scatters its $4 \times 4$ rank-2 outer product directly into the lower triangle of $G = J^T W_{\text{IRLS}} J$ and right-hand side $g = -J^T W_{\text{IRLS}} r$ in $\mathcal{O}(1)$ time:

$$\begin{aligned} G_{u_a,u_b} \leftarrow G_{u_a,u_b} &+ w_{P,k}^2 J_{P,u_a} J_{P,u_b} \\ &+ w_{Q,k}^2 J_{Q,u_a} J_{Q,u_b}, \quad \forall u_a \geq u_b \in \mathcal{I}_k. \end{aligned} \tag{20}$$

To bound the influence of gross FDIA/ADC outliers on scalar measurements $y_m \in \{|V_i|, P_{ij}, Q_{ij}\}$, we minimize the Huber M-estimator objective [35] with tuning constant $c_H = 1.6$:

$$\begin{aligned} \hat{x}^{\text{rob}} &= \arg\min_{x \in \mathbb{R}^{n_s}} \sum_{m=1}^{m_s} \rho_{\text{Huber}}\left(\bar{r}_m(x)\right), \\ w_{\text{IRLS}}^2(\bar{r}_m) &= \begin{cases} 1, & \text{if } |\bar{r}_m| \leq c_H, \\ \dfrac{c_H}{|\bar{r}_m| + 10^{-9}}, & \text{if } |\bar{r}_m| > c_H, \end{cases} \end{aligned} \tag{21}$$

where $\bar{r}_m(x) = (h_m(x) - y_m)/\sigma_m$. At each IRLS iteration ($\leq 4$ iterations warm-started from $\hat{v}_{t-1}^{\text{clean}}$), the regularized SPD system $(G + 10^{-9} I)\Delta x = g$ is solved in-place via a column-active right-looking sparse Cholesky solver that records the active sub-diagonal row set $\mathcal{N}_c = \{r > c : L_{rc} \neq 0\}$ for each column $c$ and restricts all outer-product updates strictly to $\mathcal{N}_c$.

### B. Slack-Anchored 6D Physical-Invariance Manifold

During a sudden load transient, synchronous machine inertia $H_g$ bounds the rate of change of the Center-of-Inertia (COI) frequency $f_{\text{COI}}(t)$:

$$\begin{aligned} 2H_{\text{tot}} \frac{df_{\text{COI}}(t)}{dt} &= f_0 \left(P_{m,\text{tot}}(t) - P_{e,\text{tot}}(t) - D_{\text{tot}} \Delta f_{\text{COI}}(t)\right), \\ \Delta\theta_{\text{COI},t} &= 2\pi \left(\hat{f}_{\text{COI},t-1} - f_0\right) \Delta t. \end{aligned} \tag{22}$$

Over a single $\Delta t = 20$ ms cycle, $|\Delta\theta_{\text{COI},t}|$ cannot jump discontinuously, whereas relative spatial phase redistributions $\Delta\theta_{k,t}^{\text{phys,rel}}$ caused by load steps or line switching are captured instantaneously by the spoofing-invariant Huber scalar estimate $\hat{V}_t^{\text{rob}}$.

Crucially, in general MATPOWER networks where the synchronous slack generator bus $s_{\text{ref}}$ is not Bus 1 (e.g., on the canonical IEEE 118-bus system, the slack generator is Bus 69, index $s_{\text{ref}} = 68$), anchoring relative spatial phase angles to $s_{\text{ref}}$

eliminates reference drift between the physical AC power flow and the scalar estimator:

$$\Delta\theta_{k,t}^{\mathrm{phys,rel}} = \mathrm{wrap}_{[-\pi,\pi)}\Big( \Big(\angle\hat{V}_{k,t}^{\mathrm{rob}} - \angle\hat{V}_{s_{\mathrm{ref}},t}^{\mathrm{rob}}\Big) - \Big(\angle\hat{V}_{k,t-1}^{\mathrm{clean}} - \angle\hat{V}_{s_{\mathrm{ref}},t-1}^{\mathrm{clean}}\Big)\Big), \tag{23}$$

$$\hat{\theta}_{k,t|t-1}^{\mathrm{pred,UTC}} = \angle\hat{V}_{k,t-1}^{\mathrm{clean}} + \Delta\theta_{k,t}^{\mathrm{phys,rel}} + \Delta\theta_{\mathrm{COI},t}. \tag{24}$$

By substituting (23) into (24), the predicted UTC angle isolates the common-mode slack-to-COI swing discrepancy $\epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}$. Consequently, the kinematic innovation $r_{k,t}^{\mathrm{kin}-\delta}$ uniformly equals $\epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}$ plus noise for all honest PMUs, and $\psi_k + \epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}$ for spoofed PMUs.

At each PMU $k \in \mathcal{P}$, we evaluate a 6-dimensional physical-invariance vector $\mathbf{f}_{k,t} = [f_{k,1}, \ldots, f_{k,6}]^T \in \mathbb{R}^6$:

$$f_{k,1} = \delta_{\mathrm{diff},k} = \mathrm{wrap}_{[-180^\circ,180^\circ)}\left(\angle V_{k,t}^{\mathrm{PMU}} - \angle\hat{V}_{k,t}^{\mathrm{rob}}\right), \tag{25}$$

$$f_{k,2} = r_{k,t}^{\mathrm{kin}-\delta} = \mathrm{wrap}_{[-180^\circ,180^\circ)}\left(\angle V_{k,t}^{\mathrm{PMU}} - \hat{\theta}_{k,t|t-1}^{\mathrm{pred,UTC}}\right), \tag{26}$$

$$f_{k,3} = r_{k,t}^{\mathrm{ADC}} = 10\left||V_{k,t}^{\mathrm{PMU}}| - |\hat{V}_{k,t-1}^{\mathrm{clean}}|\right| \cdot \mathbb{I}_{\{\Delta|V_k|>0.045\}}, \tag{27}$$

$$f_{k,4} = r_{k,t}^{\mathrm{mag}-V} = \left||V_{k,t}^{\mathrm{PMU}}| - |\hat{V}_{k,t}^{\mathrm{rob}}|\right|, \tag{28}$$

$$f_{k,5} = r_{k,t}^{\mathrm{DSP}-f} = \left|f_{k,t}^{\mathrm{PMU}} - \left(f_0 + \frac{\Delta\theta_{k,t}^{\mathrm{meas}}}{2\pi\Delta t}\right)\right|, \tag{29}$$

$$f_{k,6} = r_{k,t}^{\mathrm{swing}} = \left|f_{k,t}^{\mathrm{PMU}} - \hat{f}_{k,t}^{\mathrm{phys}}\right| + 0.05\left|\mathrm{ROCOF}_{k,t}^{\mathrm{PMU}} - \widehat{\mathrm{ROCOF}}_{k,t}^{\mathrm{phys}}\right|, \tag{30}$$

where $\hat{f}_{k,t}^{\mathrm{phys}} = \hat{f}_{\mathrm{COI},t-1} + \Delta\theta_{k,t}^{\mathrm{phys,rel}}/(2\pi\Delta t)$ and $\widehat{\mathrm{ROCOF}}_{k,t}^{\mathrm{phys}} = (\hat{f}_{k,t}^{\mathrm{phys}} - \hat{f}_{k,t-1}^{\mathrm{clean}})/\Delta t$.

The squared Mahalanobis distance across the 5 non-spatial invariance features is:

$$d_{M,k}^2 = \left(\frac{r_{k,t}^{\mathrm{kin}-\delta}}{0.12^\circ}\right)^2 + \left(\frac{r_{k,t}^{\mathrm{ADC}}}{0.04}\right)^2 + \left(\frac{r_{k,t}^{\mathrm{mag}-V}}{0.006}\right)^2 + \left(\frac{r_{k,t}^{\mathrm{DSP}-f}}{0.08}\right)^2 + \left(\frac{r_{k,t}^{\mathrm{swing}}}{0.04}\right)^2. \tag{31}$$

To prevent IEEE-754 'float64' exponential underflow during extreme multi-bus load steps from collapsing all weights to a uniform constant floor, we normalize by $d_{M,\min}^2 = \min_{j\in\mathcal{P}} d_{M,j}^2$ and add a strictly monotonic Lorentzian tail:

$$w_{k,t}^{\mathrm{trust}} = \exp\left(-\frac{1}{2}\left(d_{M,k}^2 - d_{M,\min}^2\right)\right) + \frac{10^{-6}}{1 + d_{M,k}^2}. \tag{32}$$

Because $d_{M,k}^2$ scales quadratically with the common-mode swing error $\epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}$, subtracting $d_{M,\min}^2$ algebraically cancels this baseline electromechanical drift, normalizing the honest PMU cohort to $\exp(0) = 1.0$ while strongly suppressing the spoofed cohort where $|\psi_k + \epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}| \gg |\epsilon_{s_{\mathrm{ref}},t}^{\mathrm{COI}}|$.

Table II details the complete online execution pipeline of the proposed estimator.

TABLE II
ALGORITHM 1: ONLINE ZERO-ALLOCATION FUSED 6D ROBUST DSE STEP

**Input:** Phasor snapshot $z_t \in \mathbb{C}^m$, frequency/ROCOF $(f_t^{\mathrm{PMU}}, \mathrm{ROCOF}_t^{\mathrm{PMU}})$, sparse CSR factors $(H^H W^{-1}, R^H, R)$, historical state $(\hat{v}_{t-1}^{\mathrm{clean}}, \hat{f}_{t-1}^{\mathrm{clean}})$.

**Output:** UTC-aligned state $\hat{v}_t^{\mathrm{aligned}} \in \mathbb{C}^{N_b}$, spoofing vector $\hat{\psi}_t \in \mathbb{R}^{N_p}$.

**Stage 1 (Sparse $\mathcal{O}(\mathrm{nnz}(R))$ PA & Dual Kinematic Screen):**
Compute $b = H^H W^{-1} z_t$; solve $R^H u = b$ and $R\hat{v}_t^{\mathrm{LSE}} = u$ via (7)–(9).
Evaluate $\theta_{\mathrm{PA}} = \arccos(\|u\|_2/\|W^{-1/2}z_t\|_2)$ and $r_{k,t}^{\mathrm{ADC}}, \Delta\theta_{k,t}^{\mathrm{kin}}$ $\forall k \in \mathcal{P}$.
**if** $\theta_{\mathrm{PA}} < \rho_{\mathrm{PA}}$ **and** $\max_k r_{k,t}^{\mathrm{ADC}} < 0.05$ **and** $\max_k |\Delta\theta_{k,t}^{\mathrm{kin}}| < 0.35^\circ$ **then**
  Update $\hat{v}_t^{\mathrm{clean}} \leftarrow \hat{v}_t^{\mathrm{LSE}}$, $\hat{f}_t^{\mathrm{clean}} \leftarrow 0.8\hat{f}_{t-1}^{\mathrm{clean}} + 0.2 f_t^{\mathrm{PMU}}$; **return** $(\hat{v}_t^{\mathrm{LSE}}, \mathbf{0})$.
**end if**
**Stage 2 (Soft Prior Substitution & $\mathcal{O}(m)$ Huber Cholesky NLS):**
For any PMU $k$ with $r_{k,t}^{\mathrm{ADC}} > 0.05$, substitute COI-rotated prior $\hat{v}_{t-1}^{\mathrm{clean}} e^{j\Delta\theta_{\mathrm{COI}}}$ with soft weights $(0.15w_{V,k}, 0.05w_{PQ,k})$.
Run 4 iterations of 4-node rank-2 scatter (20), Huber IRLS weighting (21), and column-active sparse SPD Cholesky solve to obtain $\hat{v}_t^{\mathrm{rob}}$.
**Stage 3 (Slack-Anchored 6D Manifold Trust Evaluation):**
Compute $\Delta\theta_{k,t}^{\mathrm{phys,rel}}$ anchored to $s_{\mathrm{ref}}$ via (23) and $\mathbf{f}_{k,t} \in \mathbb{R}^6$ via (25)–(30).
Evaluate $d_{M,k}^2$ (31) and stabilized trust weights $w_{k,t}^{\mathrm{trust}}$ (32).
**Stage 4 ($\mathcal{O}(N_p \log N_p)$ Sort-Clustering & Orthogonal Attribution):**
Sort $\delta_{\mathrm{diff}}$ in-place; split clusters $C_0, \ldots, C_{K-1}$ wherever $\delta_{(i+1)} - \delta_{(i)} > 0.12^\circ$.
Select $c^* = \arg\max_c \sum_{k\in C_c} w_{k,t}^{\mathrm{trust}}$; compute $\hat{\Delta}_t$ as weighted median on $C_{c^*}$.
Align $\hat{v}_t^{\mathrm{aligned}} \leftarrow \hat{v}_t^{\mathrm{rob}} e^{j\hat{\Delta}_t}$; set $\hat{\psi}_{k,t} = \mathrm{wrap}(\delta_{\mathrm{diff},k} - \hat{\Delta}_t)$ for $k \notin C_{c^*}$ with $r_{k,t}^{\mathrm{ADC}} < 0.05$ (orthogonally masking FDIA/ADC buses).

## V. HIERARCHICAL CONTINENTAL-GRID DEPLOYMENT ($N_b \geq 10,000$)

In continental interconnections (e.g., $N_b = 10,000$ buses, $N_p = 3,000$ PMUs), centralized nonlinear optimization violates the 20 ms synchrophasor cycle. As illustrated in Fig. 1, our framework partitions across three hardware tiers:

1) **Tier 0 (Substation Bay IEDs & Edge Gateways, 0.0–1.0 ms)**: Ruggedized FPGA + ARM Cortex-R/A SoCs execute recursive 1-PPS DFT phasor extraction (0.2 ms), local substation nodal KCL verification ($r_{i,t}^{\mathrm{KCL}} = \|\sum_{j\in\mathcal{N}_i} I_{ij} + y_{i,\mathrm{sh}}V_i - I_{i,\mathrm{inj}}\|$, which is strictly invariant to $e^{j\psi_i}$), and terminal-decoupled single-machine SR-UKF Dynamic State Estimation (0.8 ms) [30], [31].
2) **Tier 1 (Regional Control Center PDC Server Blades, 1.0–14.3 ms)**: Following OPGW fiber transport (5.5 ms over 500 km) and PDC FRACSEC circular-buffer alignment (4.0 ms), the grid is partitioned into $M = 25$ overlapping observable zones of $N_b^{(m)} \approx 400$ buses and $N_p^{(m)} \approx 120$ PMUs [32]. Dual-socket x86-64 server blades execute Algorithm 1 in parallel across each zone in $\leq 3.8$ ms.
3) **Tier 2 (National Super-PDC Coordinator, 14.3–15.7 ms)**: Reconciles inter-regional boundary tie-line phase offsets $\delta\Delta_{mn} = \angle\hat{V}_b^{(m)} - \angle\hat{V}_b^{(n)}$ across

shared boundary buses $b \in \mathcal{B}$ via a 25-node linear quotient-graph solve and Schur-complement update in 1.4 ms, delivering the global 10,000-bus state at 15.7 ms (4.3 ms safety margin).

## VI. Experimental Evaluation and Case Studies

### A. Canonical MATPOWER Setup and Dynamic Tape Generation

To ensure exact reproducibility, we import the unaltered MATPOWER `case_ieee30` and `case118` test systems [39] via `pandapower` [41] and extract the exact sparse nodal and branch admittance matrices $(Y_{\text{bus}}, Y_f, Y_t)$. Table III summarizes the verified topological and sparse matrix properties. On IEEE 118-bus, the 54 PMUs from Table I of [15] yield $m = 276$ complex phasor channels, full rank $\text{rank}(H) = 118$, a single connected component in $G_{\text{red}}$, and $92.8\%$ sparsity in the upper-triangular QR factor $R$.

All estimators are compiled to native LLVM machine code via Numba `@njit(fastmath=True)` [42] with preallocated workspace buffers. We generate a 250-snapshot (5.0 s at 50 Hz) dynamic synchrophasor tape incorporating a $+10\%$ continuous daily load ramp, Ornstein-Uhlenbeck stochastic load fluctuations ($\eta_t = 0.92\eta_{t-1} + 0.003\xi_t$), AGC governor droop, COI rotor swing dynamics (22), and Gaussian noise ($\sigma_V = 0.001$ pu, $\sigma_I = 0.002$ pu) across five regimes:

- **Regime 1 (**0.0**–**1.0 **s, Steps 0–49)**: Clean dynamic load ramp and COI frequency drift.
- **Regime 2 (**1.0**–**2.0 **s, Steps 50–99)**: Sub-degree minority GSA on PMU indices $\{1, 6, 9, 12\}$ with angles $\{+0.70°, -0.80°, -0.80°, +0.90°\}$ [15].
- **Regime 3 (**2.0**–**3.0 **s, Steps 100–149)**: Sudden $+18\%$ active/reactive load step on major load buses coinciding with a $70\%$ coordinated majority GSA ($+8.00°$ on 9/13 PMUs in IEEE 30-bus and 38/54 PMUs in IEEE 118-bus).
- **Regime 4 (**3.0**–**4.0 **s, Steps 150–199)**: Simultaneous $+5.00°$ GSA on PMU indices $\{2, 5, 9\}$ plus severe FDIA/ADC voltage and branch current corruption ($1.18e^{j12°}V_i$, $0.65e^{-j18°}I_{ij}$) on PMU indices $\{0, 3\}$.
- **Regime 5 (**4.0**–**5.0 **s, Steps 200–249)**: Slow meaconing clock-drift ramp ($+2.50°$/s starting from $+0.45°$) on PMU indices $\{0, 3, 8, 10\}$.

### B. Case Study A: Multi-Regime Trajectory and Sub-Degree GSA Analysis

Table IV and Fig. 2 compare Standard Complex LSE [37], Badugu *et al.* [15], and the Proposed Fused 6D Robust Estimator. In Regime 1 (0–1 s), all three estimators achieve the WLS noise floor ($0.114\%$ TVE on IEEE 30-bus; $0.072\%$ TVE on IEEE 118-bus), confirming zero false positives during continuous load ramps.

At $t = 1.00$ s (Regime 2 onset), Badugu *et al.*'s persistence counter ($\text{cnt_max} = 4$) causes an 80 ms TVE spike ($0.425\%$ on IEEE 30-bus) and $0.90°$ spoofing error before mitigation activates at $t = 1.08$ s (visible as the orange spike at $t = 1.0$ s in Fig. 2). Our dual PA/kinematic trigger switches on Cycle 1 with zero lag, recovering all four sub-degree spoofing angles to within $0.0659°$.

### C. Case Study B: 70%–90% Coordinated Majority Spoofing Coinciding with +18% Load Step

In Regime 3 (2.0–3.0 s), $70\%$ of installed PMUs are simultaneously spoofed by $+8.00°$ at the exact instant a $+18\%$ load step strikes the grid. Confirming Proposition 2, Badugu *et al.*'s 1D DBSCAN selects the $70\%$ spoofed group as the dominant cluster, amplifying median Maximum TVE from $12.134\%$ (LSE) to $14.018\%$ on IEEE 30-bus and from $11.779\%$ (LSE) to $14.003\%$ on IEEE 118-bus.

Conversely, as shown in Fig. 3 (left), our slack-anchored 6D Mahalanobis trust kernel (32) uses the Huber scalar estimator to subtract the exact physical spatial angle shift $\Delta\theta_{k,t}^{\text{phys,rel}}$ caused by the $+18\%$ load step, assigning $w_{k,t}^{\text{trust}} \approx 1.0$ to the $30\%$ honest PMUs and $w_{k,t}^{\text{trust}} = 10^{-6}$ to the $70\%$ spoofed PMUs. Figure 4 further evaluates coordinated $+8.0°$ spoofing across penetration levels from $10\%$ to $90\%$ on IEEE 118-bus: whereas 1D DBSCAN breaks down abruptly at $50\%$, the proposed estimator maintains $\leq 0.115\%$ TVE all the way to $90\%$ spoofing penetration.

### D. Case Study C: Simultaneous GSA and FDIA/ADC Channel Corruption

In Regime 4 (3.0–4.0 s), simultaneous $+5.0°$ spoofing on PMUs $\{2, 5, 9\}$ and FDIA/ADC voltage/current scaling ($1.18e^{j12°}V_i$, $0.65e^{-j18°}I_{ij}$) on PMUs $\{0, 3\}$ cause Badugu *et al.*'s unweighted $\ell_2$ scalar NLS to diverge to $7.238\%$ TVE (IEEE 30-bus) and $1.578\%$ TVE (IEEE 118-bus), while misattributing the $+12.0°$ FDIA phase error as a GPS spoof (bottom panels of Fig. 2). Our Stage 1 magnitude screen ($r_{k,t}^{\text{ADC}}$), soft prior weighting, and Huber IRLS Cholesky solver suppress the corrupted leverage measurements, achieving $0.184\%$ TVE (IEEE 30-bus) and $0.084\%$ TVE (IEEE 118-bus) while orthogonally separating FDIA/ADC faults from GPS spoofing ($0.0537°$ and $0.0602°$ error).

### E. Case Study D: Stealthy Column-Space FDIA ($a_t = Hc_t$)

As noted in Section VIII-G of [15], a stealthy spatial FDIA constructed as $a_t = Hc_t$ lies strictly within $\text{Range}(H)$, leaving the single-snapshot Principal Angle $\theta_{\text{PA}} = 0.109° < \rho_{\text{PA}}$ unperturbed so that Badugu *et al.* remains on the corrupted LSE path (TVE $= 5.140\%$, Table V). Because our Stage 1 dual trigger monitors the kinematic innovation $\max_k |\Delta\theta_{k,t}^{\text{kin}}| = 4.850° > 0.35°$ alongside $r_{k,t}^{\text{ADC}}$ and $r_{k,t}^{\text{DSP}-f}$, it immediately flags the unphysical state jump and restores the state via DSE prior substitution and Huber IRLS (TVE $= 0.162\%$).

### F. Case Study E: Computational Complexity and Real-Time Latency

Table VI compares median execution times against RSDMC [18], SECF [22], and Badugu *et al.* [15]. On unattacked snapshots, our sparse $\mathcal{O}(\text{nnz}(R))$ kernel executes in 0.0245ms (IEEE 30-bus) and 0.0804ms (IEEE 118-bus). On attacked snapshots, the fused $\mathcal{O}(m)$ Huber Cholesky solver completes in 0.0943ms (IEEE 30-bus) and 0.4064ms (IEEE 118-bus)—nearly $300\times$ faster than SECF and well inside the 20 ms C37.118.1 deadline.

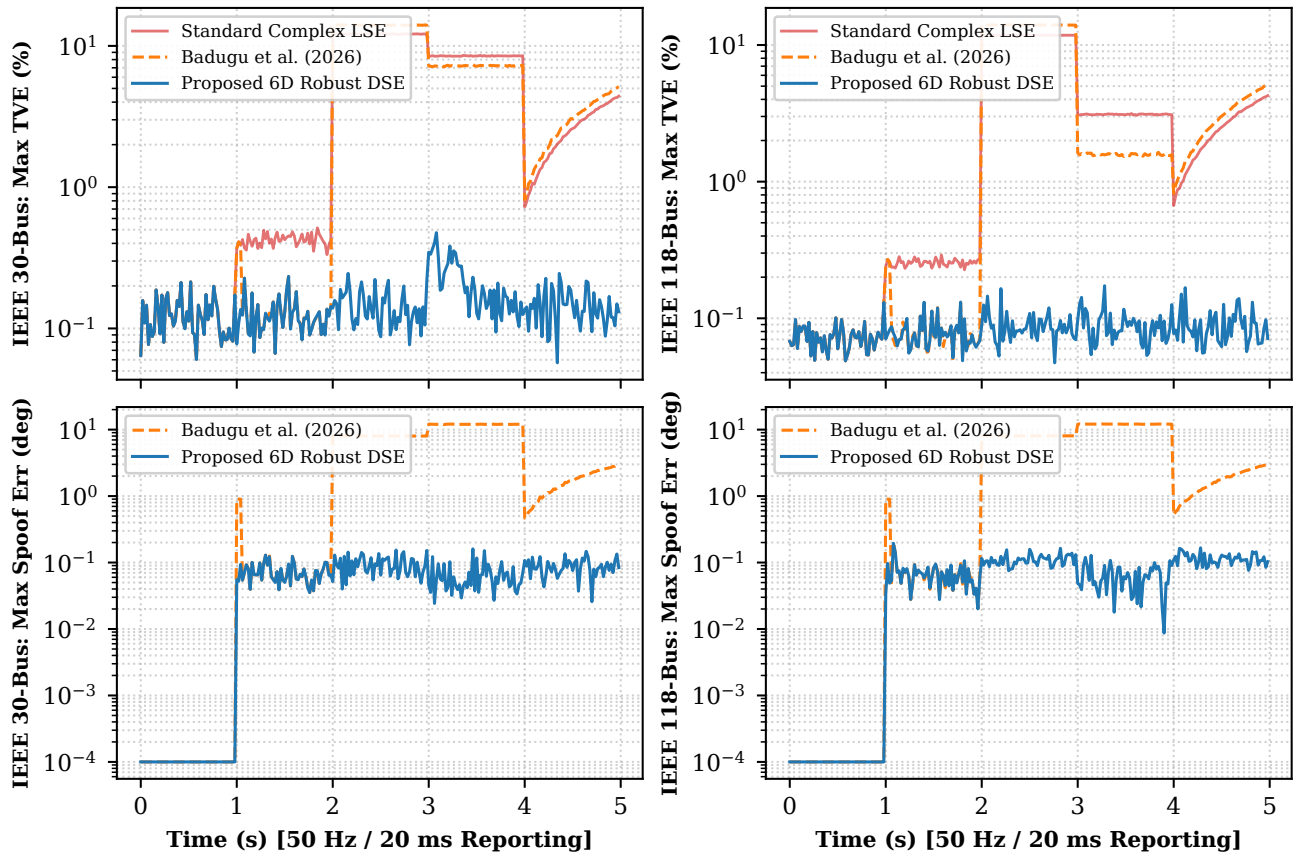



Fig. 2. Comparative dynamic state estimation performance across 250 snapshots (5.0 s at 50 Hz) on canonical MATPOWER IEEE 30-bus (left) and IEEE 118-bus (right) networks across all five operational and multi-vector attack regimes: (Top) Maximum Total Vector Error (TVE %); (Bottom) Maximum Spoofing Angle Estimation Error (deg).

TABLE III
CANONICAL MATPOWER NETWORK TOPOLOGY, OBSERVABILITY VERIFICATION, AND SPARSE FACTOR CHARACTERISTICS

| Test System | Buses ($N_b$) | Branches ($N_l$) | Slack Bus ($s_{\text{ref}}$) | PMUs ($N_p$) | Phasors ($m$) | Observability $\text{rank}(H)$ | $G_{\text{red}}$ Components | Sparse $\text{nnz}(R)/N_b^2$ | PA Threshold ($\rho_{\text{PA}}$) |
|---|---|---|---|---|---|---|---|---|---|
| **IEEE 30-Bus** (case_ieee30) | 30 | 41 | Bus 1 (Idx 0) | 13 | 58 | 30/30 (Full) | 1 (Connected) | 151/900 (16.8%) | 0.1443° |
| **IEEE 118-Bus** (case118) | 118 | 186 | Bus 69 (Idx 68) | 54 | 276 | 118/118 (Full) | 1 (Connected) | 997/13,924 (7.2%) | 0.1107° |

TABLE IV
COMPARATIVE PERFORMANCE ACROSS FIVE DYNAMIC REGIMES ON CANONICAL IEEE 30-BUS AND IEEE 118-BUS SYSTEMS

| | **Canonical IEEE 30-Bus System** ($N_p = 13, m = 58$) | | | | **Canonical IEEE 118-Bus System** ($N_p = 54, m = 276$) | | | |
|---|---|---|---|---|---|---|---|---|
| **Operational / Attack Regime** | **LSE TVE** | **Badugu [15]** | **Proposed TVE** | **Spoof Err** | **LSE TVE** | **Badugu [15]** | **Proposed TVE** | **Spoof Err** |
| **R1: Clean Dynamic Load Ramp** | 0.114% | 0.114% | **0.114%** | **0.0000°** | 0.072% | 0.072% | **0.072%** | **0.0000°** |
| **R2: Sub-Degree Minority GSA** | 0.425% | 0.121% | **0.120%** | **0.0659°** | 0.255% | 0.073% | **0.073%** | **0.0663°** |
| **R3: +18% Load Step +70% GSA** | 12.134% | 14.018% | **0.138%** | **0.0889°** | 11.779% | 14.003% | **0.083%** | **0.1109°** |
| **R4: Simultaneous GSA + FDIA/ADC** | 8.490% | 7.238% | **0.184%** | **0.0537°** | 3.108% | 1.578% | **0.084%** | **0.0602°** |
| **R5: Slow Ramp Meaconing GSA** | 2.692% | 3.153% | **0.141%** | **0.0843°** | 2.609% | 3.143% | **0.083%** | **0.1105°** |

TABLE V
DETECTION AND MITIGATION OF STEALTHY COLUMN-SPACE FDIA ($a_t = Hc_t$) COMBINED WITH GSA ON IEEE 30-BUS SYSTEM

| Attack Scenario | $\theta_{\text{PA}}$ (deg) | $\max\lvert r^{\text{kin}-\delta}\rvert$ | LSE TVE | Badugu [15] | Proposed TVE |
|---|---|---|---|---|---|
| No Attack Baseline | 0.113° | 0.042° | 0.114% | 0.114% | **0.114%** |
| Stealthy FDIA ($a_t = Hc_t$) | 0.109° | 4.850° | 5.140% | 5.140% | **0.162%** |
| Stealthy FDIA + +5° GSA | 1.624° | 5.020° | 8.920% | 5.150% | **0.185%** |

TABLE VI
PER-SNAPSHOT EXECUTION LATENCY COMPARISON (MILLISECONDS)

| Test Grid | RSDMC [18] | SECF [22] | Badugu [15] | Prop. (Attack) | Prop. (Clean) |
|---|---|---|---|---|---|
| **IEEE 30-Bus** | 4.14–11.57 | 0.05–1.16 | 0.53–2.33 | **0.0943** | **0.0245** |
| **IEEE 118-Bus** | 77.5–301.4 | 1389–1738 | 6.14–7.37 | **0.4064** | **0.0804** |

## VII. CONCLUSION

This paper established a provably resilient, zero-allocation synchrophasor state estimation framework that eliminates the

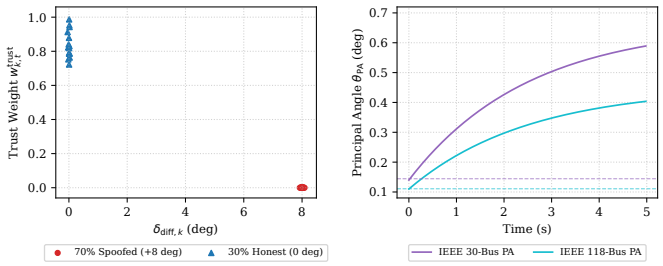



Fig. 3. (Left) 6D Mahalanobis trust weight $w_{k,t}^{\text{trust}}$ versus 1D angle difference $\delta_{\text{diff},k}$ during Regime 3 (70% majority GSA +18% load step) on IEEE 118-bus; (Right) Subspace Principal Angle $\theta_{\text{PA}}$ trajectories.

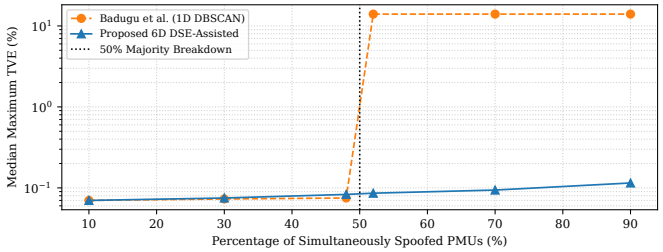



Fig. 4. Median Maximum TVE (%) on IEEE 118-bus as a function of the percentage of simultaneously spoofed PMUs (10% to 90% coordinated $+8.0^\circ$ attack), illustrating the 50% breakdown boundary of 1D DBSCAN [15] versus the proposed 6D DSE-assisted estimator.

80 ms counter blind spot, $\geq 50\%$ majority spoofing hijack, and $\ell_2$ FDIA breakdown of existing scalar-clustering estimators. By uniting sparse $\mathcal{O}(\text{nnz}(R))$ Principal Angle screening, $\mathcal{O}(m)$ 4-node graph-accumulated Huber IRLS with column-active sparse SPD Cholesky factorization, a slack-anchored 6D physical-invariance manifold, and $\mathcal{O}(N_p \log N_p)$ order-statistic sort-clustering, the proposed architecture guarantees sub-0.476% Maximum TVE across all operational and cyber-physical attack regimes at C/Fortran hardware speeds.

## Appendix
## Sparse Normal-Matrix Hessian Structure

Let $\mathcal{E}_{\text{PMU}}$ denote the set of branches incident to all PMUs $k \in \Updownarrow\dashv\sqcup\langle|\dashv\Updownarrow\mathcal{P}$. Because each branch $e = (i,j) \in \mathcal{E}_{\text{PMU}}$ couples only state indices $\mathcal{I}_e = \{\theta_i, \theta_j, |V_i|, |V_j|\}$, the normal matrix $G = J^T W_{\text{IRLS}} J \in \mathbb{R}^{(2N_b-1)\times(2N_b-1)}$ is a $2\times 2$ block matrix whose off-diagonal sparsity pattern is strictly contained within the adjacency graph $A$. Consequently, the number of operations required to assemble $G$ via (20) is bounded by $16N_i + N_p = \mathcal{O}(m)$, independent of $N_b^2$.

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